◉ OPENMTSN — CONFIDENTIAL EXECUTIVE TECHNICAL BRIEF

OpenMTSN — Executive Technical Brief

> ***Open Multi-Terrain Shared Network*** *A zero-cost, open-source disaster response network controller*

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1. Executive Summary

OpenMTSN is an intelligent network routing system designed for disaster response scenarios where traditional communication infrastructure (cell towers, fiber lines) has been destroyed or degraded. It ensures that **life-saving telemetry from field devices — drones, first-responder phones, IoT sensors — always finds a path to the command center**, even when individual network links are failing.

The system continuously monitors the health of every communication channel (5G, Satellite, Bluetooth Mesh) and **automatically switches a device to a better channel the instant the current one degrades** — with zero packet loss. If all cloud connectivity is lost, devices automatically form a local mesh network and relay data to each other until connectivity is restored.

Key differentiators: - **Zero cost** — runs entirely on free-tier cloud infrastructure and open-source software - **Cloud agnostic** — deployable to

AWS, Azure, or any Docker host with a single command - **Mathematically fault-tolerant** — routing logic validated through mutation testing that proves every decision branch is correctly covered - **Edge-resilient** — the Rust agent runs on constrained devices (Raspberry Pi, drones) with <10 MB memory footprint

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2. The Problem We Solve

📊 Architecture Diagram

```
graph LR
subgraph Before["❌ Without OpenMTSN"]
D1["🛰️ Drone sends<br/>GPS of survivors"]
X1["❌ Cell tower<br/>destroyed"]
CC1["🖥️ Command Center<br/>NEVER receives data"]
D1 -->|"Telemetry"| X1
X1 -.->|"DROPPED"| CC1
end
subgraph After["✅ With OpenMTSN"]
D2["🛰️ Drone sends<br/>GPS of survivors"]
R1["📡 5G fails →<br/>Auto-switch to<br/>🛰️ Satellite"]
CC2["🖥️ Command Center<br/>receives data in < 1s"]
D2 -->|"Telemetry"| R1
R1 -->|"Zero-drop handoff"| CC2
end
```

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During earthquakes, floods, wildfires, and conflict zones: - **70% of cell infrastructure** can be destroyed in the first 24 hours - First responders are left with **fragmented, unreliable communication** - **GPS coordinates of survivors, medical supply requests, and evacuation routes** are lost when packets drop - Existing solutions require expensive proprietary hardware and vendor lock-in

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3. How It Works — End-to-End Data Flow

📊 Architecture Diagram

```
sequenceDiagram
participant Edge as 🛰️ Edge Device<br/>(Drone / Phone / Pi)
participant Agent as 🦀 Rust Agent<br/>(On-Device Daemon)
participant MQTT as 📡 MQTT Broker<br/>(Eclipse Mosquitto)
participant API as ⚡ Routing Engine<br/>(FastAPI)
participant Redis as 💾 Redis<br/>(State Store)
participant WS as 🔌 WebSocket
participant Dash as 🖥️ Command Center<br/>(React Dashboard)
Note over Edge,Agent: Every 3 seconds
Edge->>Agent: Network interface probe<br/>(WiFi, Cellular, Bluetooth)
Agent->>Agent: Measure signal strength,<br/>packet loss, latency
Agent->>MQTT: MQTT Publish telemetry<br/>{"node":"drone-1", "signal":85,
<br/>"uplink":"5g", "gps":[12.97,77.59]}
MQTT->>API: Deliver message
Note over API: ROUTING DECISION ENGINE
API->>API: 1. Compute health score<br/>(weighted: 40% loss + 30% signal +
30% latency)
```

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4. System Architecture

📊 Architecture Diagram

```
graph TB
subgraph EDGE["🌍 EDGE LAYER — Field Devices"]
direction LR
D1["🛩️ Drone Alpha<br/>New Delhi"]
D2["📱 Field Phone<br/>Los Angeles"]
D3["🔧 Raspberry Pi<br/>Sydney"]
end
subgraph TRANSPORT["📡 TRANSPORT LAYER — Multi-Terrain Uplinks"]
direction LR
T1["📶 5G / Cellular"]
T2["🛰️ Satellite"]
T3["🔗 Bluetooth Mesh<br/>(Peer-to-Peer)"]
end
subgraph CONTROL["☁️ CONTROL PLANE — Cloud / On-Premise"]
MQTT["📨 Eclipse Mosquitto<br/>MQTT Broker<br/><i>Lightweight IoT
messaging</i>"]
```

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Layer Breakdown

LAYER	ROLE	KEY PROPERTY
Edge Layer	Field devices running the Rust agent daemon	Runs on <10 MB RAM, works offline
Transport Layer	The actual radio/network links (5G, satellite, Bluetooth)	Agent probes all available interfaces every 3 seconds
Control Plane	The "brain" — ingests telemetry, computes routing decisions	Processes thousands of telemetry messages/second
Presentation Layer	Real-time dashboard for disaster coordinators	Sub-second updates via WebSocket

5. The Routing Algorithm — The Brain of OpenMTSN

This is the core intellectual property. Every 3 seconds, each edge device sends telemetry containing its current signal strength, packet loss, and latency. The routing engine evaluates this data through a **three-stage decision pipeline**:

📊 Architecture Diagram

```
flowchart TD
START["📩 Telemetry Received<br/>from Edge Agent"] --> SCORE
SCORE["🧮 STAGE 1: Compute Health Score<br/><br/>health = 0.40 × (1 − packet_loss/100)<br/>+ 0.30 × (signal/100)<br/>+ 0.30 × (1 − latency/500)<br/><br/>Result: score ∈ [0.0, 1.0]"]
SCORE --> THRESHOLD
THRESHOLD{"🚨 STAGE 2: Threshold Check<br/><br/>packet_loss > 15%?<br/>signal_strength < 30?"}
THRESHOLD -->|"Both OK"| STABLE["✅ STABLE<br/>Stay on current uplink<br/>confidence = health_score"]
THRESHOLD -->|"Threshold Breached"| SELECT
SELECT["🔄 STAGE 3: Select Best Alternative<br/><br/>Priority Cascade:<br/>5G/Cellular fails → Satellite<br/>Satellite fails → Mesh<br/>Mesh fails → Stay on Mesh<br/>(most resilient fallback)"]
SELECT --> FAILOVER["⚠️ FAILOVER<br/>Switch to recommended uplink<br/>confidence = 1 − health_score"]
STABLE --> STORE["💾 Store in Redis<br/>Broadcast to Dashboard"]
FAILOVER --> STORE
```

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Why These Weights?

FACTOR	WEIGHT	RATIONALE
Packet Loss	**40%** (highest)	Dropped packets directly mean lost survivor GPS coordinates. This is the most critical metric for life-saving data.
Signal Strength	**30%**	Predicts imminent connection failure. Low signal today means dropped connection in minutes.
Latency	**30%**	High latency degrades real-time coordination but doesn't lose data. Important but less critical than loss.

Failover Priority — Geographic Diversity Strategy

📊 Architecture Diagram

```
graph LR
FG["📶 5G / Cellular<br/>(Terrestrial)"] -->|"If degraded"| SAT["🛰️ Satellite<br/>(Space-based)"]
SAT -->|"If degraded"| MESH["🔗 Mesh<br/>(Peer-to-peer)"]
MESH -->|"All degraded"| MESH
```

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When terrestrial links (5G/cellular) fail — typically due to tower destruction — the system **deliberately selects satellite** rather than mesh, because satellite provides **geographic diversity** (the failure that destroyed the cell tower cannot affect a satellite link). Mesh is the last resort because while it guarantees local peer connectivity, its range is limited to ~100 metres.

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6. Mesh Fallback — The Safety Net

📊 Architecture Diagram

```
sequenceDiagram
participant A as 🚁 Drone Alpha
participant Broker as 📨 MQTT Broker
participant B as 📱 Phone Beta
participant C as 🍓 Pi Gamma
Note over A,Broker: Normal Operation
A->>Broker: MQTT Publish ✅
A->>Broker: MQTT Publish ✅
Note over A,Broker: Cloud Connectivity Lost
A-xBroker: MQTT Publish ❌ (Failure 1)
A-xBroker: MQTT Publish ❌ (Failure 2)
A-xBroker: MQTT Publish ❌ (Failure 3)
Note over A: 3 consecutive failures →<br/>ACTIVATE MESH MODE
A-->>B: UDP Broadcast<br/>on port 9876
A-->>C: UDP Broadcast<br/>on port 9876
```

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The agent automatically switches to **UDP broadcast mesh mode** after 3 consecutive MQTT failures. Nearby OpenMTSN agents listen on port 9876 and relay the data when their own connectivity is restored. **No data is ever permanently lost.**

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7. CI/CD Pipeline Workflow

📊 Architecture Diagram

```
graph LR
subgraph PR["Pull Request"]
LINT["🔍 Lint<br/>ruff + black + prettier"]
TEST["🧪 Test<br/>pytest (27 tests)"]
MUT["🧬 Mutation Test<br/>mutmut"]
RUST["🦀 Rust Tests<br/>cargo test"]
LINT --> TEST --> MUT
LINT --> RUST
end
subgraph MAIN["Push to main"]
BUILD["🐳 Docker Buildx<br/>AMD64 + ARM64"]
PUSH["📦 Push to GHCR<br/>ghcr.io/openmtsn/*"]
DEPLOY["🚀 Zero-Downtime Deploy<br/>SSH → docker compose pull<br/>→ rolling restart<br/>→ health check<br/>→ rollback if failed"]
BUILD --> PUSH --> DEPLOY
end
```

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8. Multi-Cloud Deployment Model

📊 Architecture Diagram

```
graph TB
subgraph AWS["☁️ AWS Free Tier"]
EC2["EC2 t2.micro<br/>(1 vCPU, 1 GB RAM)"]
EBS["EBS 20 GB gp3<br/>(Encrypted)"]
SG["Security Group<br/>HTTP/HTTPS/MQTT/SSH"]
EC2 --- EBS
EC2 --- SG
end
subgraph Azure["☁️ Azure Free Tier"]
APP["App Service F1<br/>(Free, 1 GB RAM)"]
IOT["IoT Hub F1<br/>(Free, 8000 msgs/day)"]
REDIS["Redis Basic C0<br/>(250 MB)"]
APP --- IOT
APP --- REDIS
end
```

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Total recurring cost: $0 — both deployment options use exclusively free-tier resources, making OpenMTSN accessible to NGOs with zero budget.

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9. Complete Technology Stack

Core Application

COMPONENT	TECHNOLOGY	VERSION	PURPOSE
Control Plane API	**Python** + **FastAPI**	3.12 / 0.111	Async HTTP + WebSocket server for telemetry ingestion and routing
Data Validation	**Pydantic v2**	2.7	Strict runtime type checking for all API contracts
Configuration	**Pydantic Settings**	2.3	Environment-variable-driven config with validation
State Store	**Redis**	7.x	In-memory topology storage with TTL-based stale node eviction
Redis Client	**redis-py** (async)	5.0	Connection pooling with `hiredis` C parser for speed
Edge Agent	**Rust**	1.78	Memory-safe, zero-cost abstractions, <10 MB runtime footprint
Async Runtime	**Tokio**	1.x	Rust's production async runtime for concurrent I/O
MQTT Client	**rumqttc**	0.24	Lightweight, async MQTT v5 client for IoT telemetry
System Monitor	**sysinfo**	0.30	Cross-platform network interface and system resource probing
Message Broker	**Eclipse Mosquitto**	2.x	Lightweight MQTT broker purpose-built for IoT

Frontend Dashboard

COMPONENT	TECHNOLOGY	VERSION	PURPOSE
UI Framework	**React**	18.3	Component-based UI with concurrent rendering
Type System	**TypeScript**	5.5	Compile-time type safety for the entire frontend
Build Tool	**Vite**	5.3	Sub-second HMR, optimized production builds
Mapping	**Leaflet** + **react-leaflet**	1.9 / 4.2	Interactive topographical map with dark CARTO tiles
Real-Time	**WebSocket** (native)	—	Sub-second topology push from API to dashboard
Typography	**Inter** + **JetBrains Mono**	—	Premium sans-serif + monospace for metrics

DevOps & Infrastructure

COMPONENT	TECHNOLOGY	VERSION	PURPOSE
Containerisation	**Docker**	24+	Multi-stage builds, non-root containers
Orchestration	**Docker Compose**	2.27	Single-command 7-service sandbox
CI/CD	**GitHub Actions**	—	Automated lint, test, build, and deploy pipeline
Multi-Arch Build	**Docker Buildx** + **QEMU**	—	AMD64 + ARM64 images for x86 servers and Raspberry Pi
Container Registry	**GHCR** (ghcr.io)	—	Free, integrated image hosting
IaC (AWS)	**Terraform**	1.5+	EC2 t2.micro, security groups, Docker bootstrap
IaC (Azure)	**Terraform**	1.5+	App Service F1, IoT Hub F1, Redis Basic
DNS / SSL / DDoS	**Cloudflare**	Free tier	Edge caching, auto SSL, L3/L4/L7 DDoS mitigation
Chaos Testing	**tc / netem** (Linux kernel)	—	Packet loss, latency, and corruption injection

Code Quality & Testing

TOOL	PURPOSE
pytest + **pytest-asyncio**	Async unit and integration testing (27 tests)
httpx (async)	In-process ASGI test client for API testing
fakeredis	In-memory Redis mock for deterministic tests
mutmut	Mutation testing — proves every routing decision branch is covered
cargo test	Rust unit tests for telemetry serialisation and network probing
ruff	Fast Python linter (replaces flake8 + isort)
black	Deterministic Python code formatter
prettier	JavaScript/TypeScript/CSS code formatter
pre-commit	Git hook framework enforcing code quality on every commit

10. Testing & Fault Tolerance Strategy

📊 Architecture Diagram

```
graph BT
subgraph TESTS["Testing Pyramid"]
UT["🧪 Unit Tests (17)<br/>Routing engine logic,<br/>health scoring,
thresholds"]
IT["🔌 Integration Tests (10)<br/>Full API flow with<br/>fakeredis
backend"]
MT["🧬 Mutation Tests<br/>Systematically breaks routing code<br/>to prove
tests catch every bug"]
CT["💥 Chaos Tests<br/>80% packet loss injected<br/>into live containers"]
end
UT --> IT --> MT --> CT
```

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Mutation testing is the key differentiator. While traditional tests verify "does the code work?", mutation testing answers **"would the tests catch it if the code was wrong?"** It systematically introduces small changes (mutations) to the routing engine — flipping `>` to `>=`, changing `15.0` to `16.0`, etc. — and verifies that at least one test fails for every mutation. This mathematically guarantees that the routing logic has **zero undetected edge-case bugs**.

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11. Real-World Use Cases

SCENARIO	HOW OPENMTSN HELPS
Earthquake Response	Drones survey collapsed buildings. When cell towers fall, drones auto-switch to satellite uplink. Command center receives survivor GPS coordinates without interruption.
Flood Zone Coordination	Field phones carried by rescue teams in flooded areas with no cell coverage form a Bluetooth mesh. Data relays hop through peers until one team member reaches satellite range.
Wildfire Monitoring	Raspberry Pi sensors at fire perimeters transmit temperature and wind data. As fire destroys nearby infrastructure, the system seamlessly routes data through remaining satellite links.
Refugee Camp Connectivity	Low-cost devices provide basic connectivity in areas with zero infrastructure. The mesh network allows camp-wide communication without any external connectivity.

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12. One-Command Demo

Anyone can run the entire system locally in under 60 seconds:

```
git clone https://github.com/openmtsn/openmtsn.git
cd openmtsn
docker compose up -d
```

This launches **7 services**: MQTT broker, Redis, FastAPI API, React dashboard, and 3 simulated edge agents transmitting telemetry from New Delhi, Los Angeles, and Sydney.

- **Dashboard**: http://localhost:5173 — live dark map with color-coded nodes
- **API Docs**: http://localhost:8000/docs — interactive Swagger UI

- **Chaos Test**: `./simulator/chaos.sh full` — inject 80% packet loss and watch the failover in real-time

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13. Key Engineering Decisions

DECISION	RATIONALE
Rust for the edge agent (not Python/Go)	Memory safety without garbage collection pauses. Critical for constrained devices where a segfault means a drone loses contact. <10 MB runtime footprint.
MQTT over HTTP for telemetry	MQTT is purpose-built for IoT: persistent connections, QoS guarantees, 10x less overhead than HTTP. Designed for unreliable networks.
Redis over PostgreSQL for topology	Sub-millisecond reads. TTL-based auto-eviction of stale nodes. The topology is transient state, not permanent data — Redis is the right tool.
Weighted scoring over rule-based routing	A composite score provides nuanced decisions. Pure threshold rules would miss scenarios where multiple metrics are slightly degraded simultaneously.
UDP broadcast for mesh fallback	When cloud connectivity is lost, TCP connection establishment is unreliable. UDP broadcast is connectionless and reaches all peers on the subnet instantly.
Multi-arch Docker builds	OpenMTSN runs on both x86 cloud servers and ARM64 Raspberry Pis. A single `docker compose up` works identically on both architectures.

This document covers the complete OpenMTSN system. Every component described above is fully implemented and ready for demonstration.

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